Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
C		2403861

Strategies to Match Swings: Based on Momentum Turning Prediction with XGBoost Regression

Summary

The 2023 Wimbledon tennis match is deeply concerned by fans, which pays particular attention to the battle for the men's championship. In this match, the power of **momentum** was on display. This paper aims to study the role of forces in tennis matches, quantify players' performance at a specific time through data analysis, explore random effects, find out the most relevant indicators, and make suggestions for players and coaches.

As for question 1, to capture the flow of play as points occur and apply it to **"2023-wimbledon-1701"** match, we believe that player momentum can reflect player performance, **Multiple Linear Regression** was used based on five correlated attributes (**Positive:** Scoring advantage, Games won, Server, Breakpoint; **Negative:** Unforced errors) carry out the analysis momentum. The **entropy method** is used to obtain the weight of the above attributes of each player in the match. Depending on these, established the linear equation for calculating the provided time, the momentum of each period of the athlete is calculated and the visualization curve is drawn. **As regards the performance evaluation**, we have categorized the evaluation range into **six levels** based on the aforementioned results. The player's performance score is determined by the corresponding value interval of the calculated result. In this paper, we conduct a comprehensive analysis of both players' performances in the match.

Question 2 is to prove whether the momentum influence is random for a player in swings in play and successful runs. We use the **Sliding window model** to obtain the turning point of the player's momentum transformation trend in the corresponding time window, and then use the **Run testing** to detect each momentum value and turning point is random or not. According to the Z-statistic and p-value, the influence is **not random and "momentum" plays an important role in the match.**

Question 3 is divided into two parts. In Part 1, in order to confirm the indicators that determine the transition of the competition process, we use the **XGBoost regression model** under the comparison of multiple models. Using the data of the first question, taking player 1 as the benchmark, the ratio of momentum transformation influence is extracted from each attribute of the game, and the corresponding feature importance score is obtained by using the model. The accuracy of the model is up to **98%**. After that, the descending order of feature importance is plotted, and it is found that **Serve, Match Points, Initiative** and other factors are most related to the fluctuations in the competition. In Part 2, we give players advice on these factors to deal with the new game.

In question 4, we choose **"2023-wimbledon-1301"** match to test model, and the obtained results prove the stability and accuracy of our model. Future model improvement may need to consider the **field factor**, **the player's own ability** and other circumstances. We can be confident that our model will be able to predict other matches by corresponding data, and we have high confidence in its generalization.

In summary, the model developed shows good performance. Its analysis and prediction can be used as a tool for momentum judgment of game scores and players to formulate countermeasures.

Keywords: **Linear Regression, Entropy method, Sliding window model, Run test, XGBoost**

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1 Introduction

1.1 Background

In the 2023 Wimbledon men's final, 20-year-old Spanish rising star Carlos Alcaraz defeated 36-year-old Novak Djokovic[**8**]. It is known that the scoring rules of men's tennis matches are three out of five sets for the winner. According to the information obtained, we can know the specific situation of the court is as follows: In this match, the veteran Djokovic dominated the first set 6-1. However, he lost 6-7 to Alcaraz in a tight second set. After that, in the third set, Alcaraz easily won 6-1. In the fourth set, the tide of the match changed and Djokovic won 6-3. Finally, in the fifth set, although Djokovic initially extended his advantage, the tide changed again, and Alcaraz won the set 6-4 and the match. Based on the data from this game, it seems that the dominant side sometimes has huge fluctuations, and we focused on the cause of this phenomenon - "momentum".

1.2 Introduction of Momentum

One dictionary definition of momentum is "strength or force gained by motion or by a series of events." In sports, a team or player may feel they have the momentum, or "strength/force" during a match or game. Often, it may simply be the fraction of points won by a player over a short period. It is also possible to record the number of games in which a player consistently scores or wins, in particular serves or counter-serves, thus specifying a momentum score. Momentum is an abstract term that can be quantified as a deviation from a player's average or expected performance during a match. Momentum can be quantified by calculating a player's scoring advantage over an opponent in a specific time window.

1.3 Restatement of the Problem and Our Analyses

1.3.1 Specific Problems to Solve

- **Question 1 :** Capture the flow of match points as they occur, identifying specific periods when a player is better in a match and to what extent the player is better. In tennis, the serving side has a much higher probability of winning the match point and the match. To solve this problem, we set up models to provide visualizations of the game flow.

- **Question 2 :** Proving that the fluctuations and players in the game are not random, evaluate the role of momentum in the game. To solve this problem, we modeled it using relevant models to find the relationship between momentum and performance.

- **Question 3 :** Indicators to determine the flow of a game from one player to another should be as follows:

  (1)Using the data provided for at least one match, predict fluctuations in the match and identify the most relevant factors;

  (2)Considering the difference in momentum fluctuations in past matches, players are advised to choose different opponents in new matches

  To accomplish this goal, we use the given data to model, and according to the model results, identify the factors and give the appropriate conclusions.

- **Question 4 :** Using the model to predict volatility for one or more other matches, and if the model does not perform well, the factors that need to be included in the future model are found, and the generality of the model to other matches is tested. To solve this problem, we will use appropriate methods to test the sensitivity and applicability of the model.

1.3.2 Analyses of the Problems

To solve the above problems, we will proceed as follows:

- **Preprocessing data**. To reduce the influence of abnormal data and improve the effect of subsequent analysis, it is necessary to preprocess the data.
- **Stating assumptions**. By stating our assumptions, we will avoid uncontrollable minor factors and simplify the modeling part by the next process.
- **Making notations**. To clarify and better illustrate our model, we will provide some important notations.
- **Establishing models, making analyses and finding solutions**. We model the problem, analyze the data, and use the model to solve the appropriate problem.
- **Evaluating our models**. We find out the advantages and the disadvantages of our models through the evaluation.
- **Further discussion**. We discuss several interesting features that we have found in the data set and make advice.

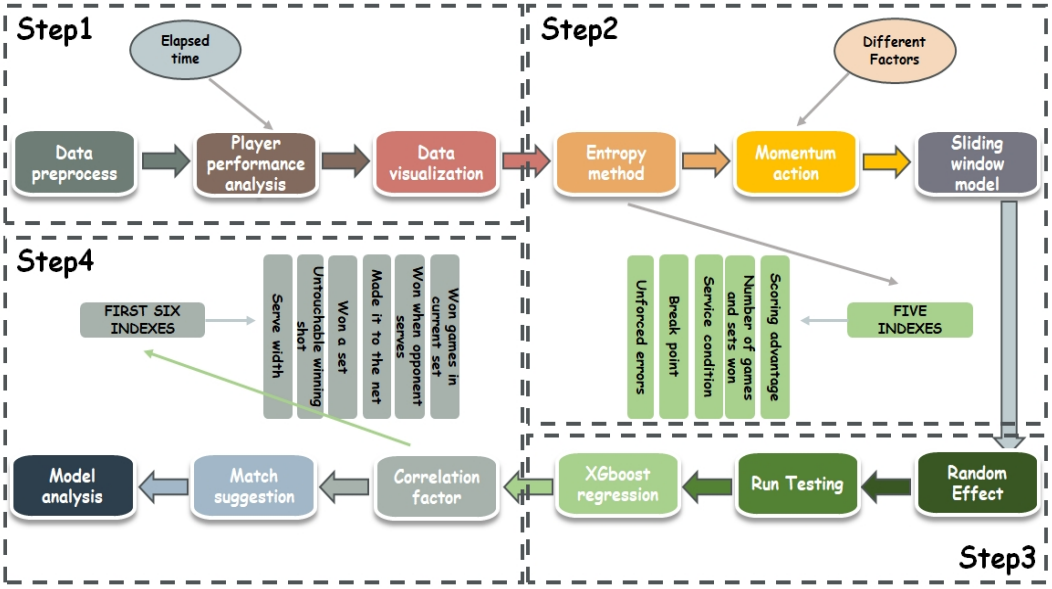


Figure 1: The Flow Chart in This Paper

2 Data preprocessing

In order to improve the quality, the accuracy and the effectiveness of data, we adopt data cleaning technique.

- **Convert:** Changing the category data into quantitative value. for example labels the D and ND values in the class data return_depth as 1,2. Thus, it is convenient for data processing and comparison.
- **Select:** Matches with more than 250 data points were selected for analysis.
- **Outliers:** Outliers 1,2,3,4,5,6,7 in p1_score and p2_score attributes are removed.

3 Assumptions and Occasions

We aim to simplify our models and ensure the accuracy of our analyses. Therefore, we make the following assumptions with adequate reasons:

- **We assume that the physical fitness of the athletes does not affect their performance.** When analyzing the corresponding performance of the players in a specific period of the match, the influence of the players' own physical quality and tennis level on their performance and momentum is approximately consistent.
- **Data after preprocessing is enough to support our models**. The amount of data available for each competition ranges from 150 to more than 300, depending on the dataset provided. The data after data preprocessing is sufficient for us to analyze and build our model.
- **The field environment does not affect the momentum and performance of the players**. It is assumed that athletes change the venue and the surrounding environment of the stadium during the competition, and the weather on that day will not affect the performance of athletes. In order to model and analyze the problem.

4 Notations

Symbols	Description
R_j	Correlation coefficient between indicators
W_j	The objective weight of the j_{th} index
M	Momentum value based on differnet match attributes
w_n	The value of the attributes (shown in pie figure 3), n = 1,2,3,...,7
K	Represents the number of parameters in the model
$\widehat{y_i}^{(t)}$	The predicted value of the t th model
Ω	XGBoost model complexity function term

where we define the main parameters while specific define of those parameters will be given later.

5 Model Establishment and Problem Solutions

5.1 Question 1

In our model, we aim to achieve two goals: build a model that can get the performance indicators of the athletes in a provided period based on the scores of the athletes in different competitions, and judge the degree of the performance of the athletes.

The Entropy weighting method is used to calculate and analyze the weight of the athlete's performance. By using the linear equation, we can approximate the performance score of the athlete at different times, which is called momentum.

5.1.1 Model I : Judge the degree of the performance of players by entropy weight

- **Analyze the data**

  Momentum can be defined as a measure of a player's current performance, the deviation from the average or expected performance during the game. That is, momentum can be quantified by calculating factors such as a player's scoring advantage over his opponent in a specific time frame. Let's define a momentum function that uses as much of the data as possible to calculate the momentum of players 1, and 2 for a specific time frame.

  We chose match_id **"2023-wimbledon-1701"** to analyze the performance of **Carlos Alcaraz** and **Novak Djokovic**. For analysis purposes, we changed the elapsed_time attribute to minutes to show the flow of the race.

- **Determine attributes of Momentum**

  There are five attributes that we consider to have an impact on momentum:

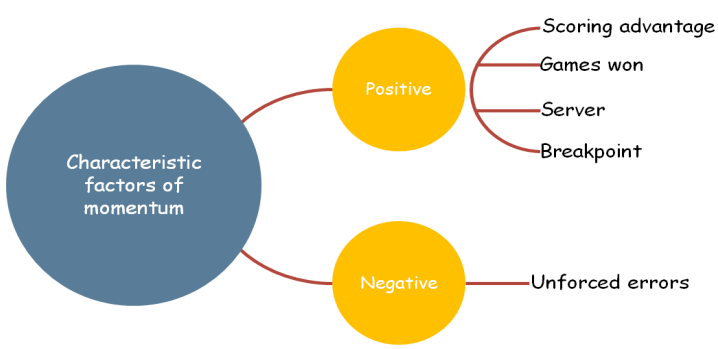


Figure 2: attributes that impact momentum

(1) Scoring advantage: Compare the score difference between two players in a specific time frame. The larger the gap, the higher the score, has the advantage.

(2) Games won: Compare the number of games won by each player over a time window. Higher numbers imply more momentum.

(3) Server: In tennis, the server has a greater advantage. We consider the fraction of points won by the server and whether the server is able to save serve (i.e., win its own service game).

(4) Breakpoint: Breakpoint indicates a player's ability to win the game when opponent serves.

(5) Unforced errors: Fewer unforced errors can be seen as better performance, and vice versa, poor performance.

- **Property Description**

  In the case of player 1, we set the following attributes to be associated with momentum:

  (1) p1_points correspond to the scoring advantage: Which indicates the difference in scoring between this player and another player in a provided period of time.

  (2) p1_serve corresponds to serve: Quantifies the player's additional advantage in scoring as the server.

  (3) p1_break_points_won corresponds to breakpoint: denotes the number of breakpoints won by player 1 in the time window and reflects the player's ability to win a match on the opponent's serve.

  (4) p1_unf_errors corresponds to unforced errors: A player's error rate is measured by the total number of unforced errors in the period (taking a negative value), with fewer unforced errors indicating better performance.

  (5) p1_sets_won and p1_games_won correspond to games won: the number of sets and games won by a player in a given time.

  The above p1_serve, p1_sets_won and p1_games_won are re-computed by matching the corresponding values in the data server, p1_sets and p1_games

  In addition, we added a personal ability assessment. p1_winners corresponding **ability score**: The total number of points scored by player 1 in the time window, indicating his initiative.

- **Develop our model**

  We introduce **entropy weight method** [**1**] attribute to assign weight value. For an index, entropy value can be used to judge the degree of dispersion of an index. The smaller the information entropy value, the greater the degree of dispersion of the index, and the greater the influence (i.e. weight) of the index on the comprehensive evaluation. If all the values of an index are equal, the index will not play a role in the comprehensive evaluation.

  The steps are as follows:

  Each factor was normalized according to the number of each option. Since there are "positive indicators" and "negative indicators" in the platform, these two types of data will be preprocessed separately.

  Positive index equation:

$$x_{ij}{}' = \frac{X_{ij} - min(X_{1j}, X_{nj}, , X_{nj})}{max(X_{1j}, X_{nj}, , X_{nj}) - min((X_{1j}, X_{nj}, , X_{nj}))} \tag{1}$$

Negative index equation:

$$x_{ij}{}' = \frac{max(X_{1j}, X_{nj}, , X_{nj}) - X_i j}{max(X_{1j}, X_{nj}, , X_{nj}) - min((X_{1j}, X_{nj}, , X_{nj}))} \tag{2}$$

Calculate the entropy value of the J index:

$$e_j = -k \sum_{i=1}^{n} ln(p_i j), j = 1, ..., m \tag{3}$$

$where \quad k = 1/ln(n) > 0(e_j > 0)$

Computing information entropy redundancy (difference) :

$$d_j = 1 - e_j \tag{4}$$

Calculate the composite score for each sample:

$$w_j = \frac{d_j}{\sum_{j=1}^{m} d_j} \tag{5}$$

Calculate the composite score for each sample:

$$s_i = \sum_{j=1}^{m} w_j x_{ij}, i = 1, ..., n \tag{6}$$

We put the given data into equations to get the weight proportion Figure by using Python.

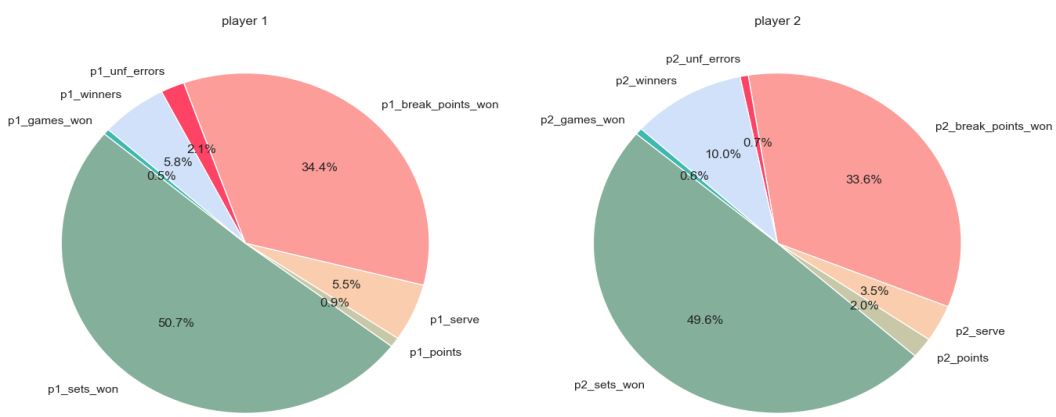


Figure 3: weight proportion of the attributes

We use the weight proportion to gain a momentum liner equation based on attributes. Combined with momentum value M and given time elapsed since start of first point to start of current point (elapsed_time) property, we obtain the following figure by Python:

For player 1 at one point:

$$M_1 = 0.507w_1 + 0.009w_2 + 0.054w_3 + 0.344w_4 + 0.021w_5 + 0.058w_6 + 0.005w_7 \quad (7)$$

For player 2 at one point:

$$M_2 = 0.496w_1 + 0.020w_2 + 0.035w_3 + 0.336w_4 + 0.007w_5 + 0.100w_6 + 0.006w_7 \quad (8)$$

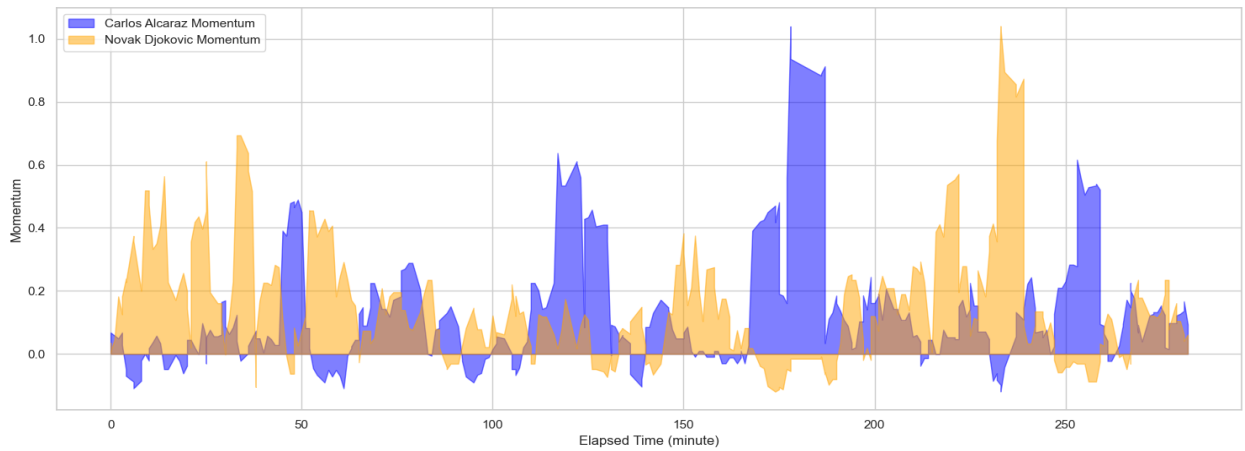


Figure 4: Momentum Change with Area Chart Throughout the Match

5.1.2 Question1 Conclusion

- **Momentum & Performance level**

  We set an athlete's performance score interval as shown in the table below:

Table 1: Momentum & Performance level Table

interval	≤ 0.0	$0.0 \sim 0.2$	$0.2 \sim 0.4$	$0.4 \sim 0.6$	$0.6 \sim 0.8$	≥ 0.8
Momentum& Performance	Terrible	Normal	Good	Great	Very Good	Excellent

We divided the elapsed time into six groups, each with a ratio of 1/6, setting the first two groups of time as the early period of the game, the middle two periods as the middle period of the game, and the last two periods of time as the late period of the game.

In addition, we also divided performance and momentum into 6 grades as shown in the above table. Through the above flow chart, we can get the performance of the two athletes in the competition at different time periods.

- **Analysis of the Match**

  In this model, we divide the time from the beginning into groups of every 50 minutes.

(1) In the first 50 minutes, Novak's momentum is very high from the beginning and can even reach very good level, and it is not until near the end of the first 50 minutes that Novak's momentum shows a significant decrease. During the same period, Carlos's momentum repeatedly rose and fell, and the absolute value of momentum at its highest and lowest was not very large, while Carlos's momentum showed a significant rise to good when the first 50 minutes was about to end and Novak's momentum dropped significantly.

(2) In the second 50 minutes, Novak's momentum was higher at the beginning, reaching good, but then there was a relative drop in momentum, but momentum was still in the normal range overall. carlos's momentum was lower at the beginning and end of the second 50 minutes, in the terrible range Carlos' momentum is lower at the beginning and end of the second 50 minutes, falling into the terrible range, but in the middle of the period momentum is comparable to Novak's momentum and is in the normal range.

(3) In the third 50 minutes, Novak's momentum is lower, varying between normal and terrible. However, in this 50 minutes, Carlos has a lower momentum at the beginning and end, and a higher momentum in the middle, even reaching very good.

(4) In the fourth 50-minute period, Novak maintains a high momentum at the beginning and the end and is able to achieve good, but in the middle the momentum is terrible, and Carlos performs similarly to the previous 50-minute period, with a low momentum at the beginning and the end, and a very high momentum in the middle and is able to achieve excellent.

(5) In the fifth 50-minute period, Novak's momentum gradually increased and was also able to reach excellent, but Carlos's momentum dropped back and generally stayed in the normal range.

(6) In the sixth 50 minutes, Novak's momentum gradually increased from terrible to normal, and Carlos' momentum gradually decreased from great to normal.

Overall, Novak's momentum is higher at the beginning of the match, then gradually decreases and peaks after 200 minutes, while Carlos's momentum is at a higher level between 100 and 200 minutes into the match. Novak's momentum is higher at the beginning of the match and peaks later than Carlos'.

5.2 Question 2

5.2.1 Model II : Explore whether there is a random effect of momentum on swings and success

- **Analysis**

  The **sliding window method**[**2**] is a commonly used time series analysis technique to detect turning points or abrupt changes in data. It identifies changes in the data by moving a fixed-size window over the time series and applying some statistical method at each window position.

- **Preparation** Here are the basic steps of the sliding window method:

  (1) Define Window size: Determines the size of the window to move over the time series. The choice of window size depends on the characteristics of the data and the requirements

of the problem. Smaller Windows may result in greater sensitivity to noise, while larger Windows may obscure the true turning point.

(2) Moving window: Starting from the starting position of the time series, the window is moved to the next position until the end of the series is reached. At each window position, the data within the window were processed as subsequence.

(3)Applying statistical methods: For a subsequence at each window position, some statistical method is applied to calculate its eigenvalues or statistics. Common methods include calculating mean, variance, difference, etc.

(3)Detect turning point: According to the defined turning point criteria, determine whether there is a turning point in the subsequence at the current window position. The definition of turning points can be based on thresholds, outlier detection, mutation detection, etc.

The mean of the data in the window:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{9}$$

Variance of the data in the window:

$$\sigma = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2 \tag{10}$$

The difference of the data in the window:

$$\Delta x_i = x_i - x_{i-1} \tag{11}$$

- **Momentum Turning Point Prediction**

  We used a sliding window approach to detect turning points in the time-series data, and we graphically displayed the changes in momentum of the two players and these turning points. **Twenty was chosen as the window size and 0.1 as the change rate** threshold in order to ensure that momentum changes in athletes could be detected within a short period of time.

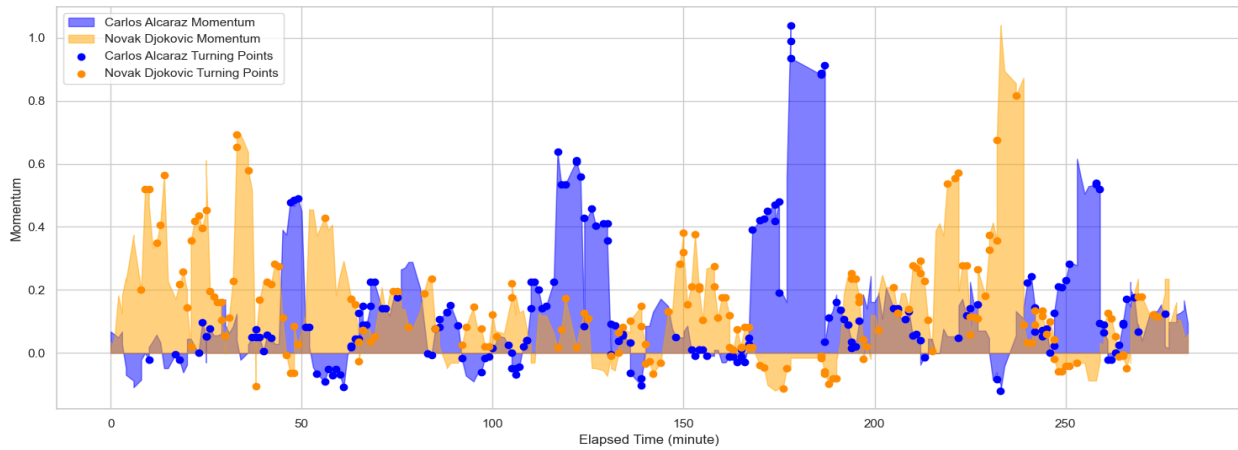


Figure 5: Momentum Change with Turning point

In the code, we compute the momentum for each row and assign the result (the turning point between the two) back to the two columns in the original dataset. The index position of the momentum turning point is identified using the Sliding window algorithm. Mark the index position of the turning point as 1 and the others as 0.

It can be seen that when the trend of the momentum value changes, the model can be correctly displayed at the change point, indicating that the model has a high accuracy.

5.2.2 Model III: Randomness testing

- **Run testing**

  Since we have transformed the data into a binary sequence, we can apply a **run test** that is suitable for determining the randomness and non-randomness of the sequence. [**3**] Run test does not need to make any assumptions on the data, does not need to know the probability distribution of the data, and is relatively simple and easy to implement.

  We want to calculate the expected value and variance of the number of runs in the sequence, and then compare it with the actual number of runs to produce a statistic (the Z-statistic) to determine whether the sequence is significantly non-random. We run the momentum series and the turning point series separately and evaluate the significance level of the series based on the p-value.

  The expected number of runs is calculated by counting the frequencies of the two types of elements in the sequence:

$$E = \frac{2n_1n_2}{n_1 + n_2} + 1 \tag{12}$$

  Where n_1 is the number of elements labeled as 1 and n_2 as 0.

Estimated variance using the expected number of runs:

$$V = \frac{(E-1)(E-2)}{n_1 + n_2 - 1} \tag{13}$$

Based on the number of runs observed, the number of runs expected, and the variance, the Z-statistic was calculated:

$$Z = \frac{observed_runs - E}{\sqrt{V}} \tag{14}$$

P-values were calculated from the Z-statistic and used to determine whether the number of runs observed was significant. A two-tailed test was used here, and the p-value was calculated using the formula:

$$p = 2 \times (1 - CDF(Z)) \tag{15}$$

CDF [**4**] represents the cumulative distribution function of the standard normal distribution.

We get the following table and conclusion through python:

Table 2: Runs Test Table

Test object	Sequence	Z statistics	P values
Momentum Test	player1	-14.0281	0.000
Momentum Test	player2	-12.6032	0.000
Turning Points Test	player1	-13.2669	0.000
Turning Points Test	player2	-13.4168	0.000

The Z-statistic indicates the degree of deviation between the observed number of runs and the expected number of runs, with larger absolute values indicating greater deviation. The p-value represents the probability that the observed number of runs occurs in a random sequence, and a small p-value indicates that the observed number of runs is very unlikely to occur in a random sequence.

5.2.3 Question2 Conclusion

In judging the randomness of the sequence, it is found that in the above tests, the Z-statistic deviates from 0 and has a large deviation and a small p value (usually less than 0.05), which proves that the observed sequence has significant characteristics in terms of run length and does not conform to the randomness assumption. **Thus the swing and success of a player in a match are not random, but have a certain positive correlation with Momentum.**

5.3 Question 3 —— Predict fluctuations and analyze related factors

5.3.1 Model IV: Anticipate fluctuations in the race

When the flow of the game changes from one player to another, it is very likely that the momentum of that player will be better than that of the other player, and its momentum will increase.

We will predict these fluctuations in the game from the increase in the player's own momentum and the difference in the momentum of another player. We take player 1 as the baseline.

Here, we consider the lag difference of momentum values as a feature to capture the trend of the game dynamics. The **XGBoost algorithm** [**2**] was used to train the model on the training set.

- **XGBoost Regression Introduction**

  XGBoost is short for "Extreme Gradient Boosting". XGBoost algorithm is a kind of synthetic algorithm that combines basis functions and weights to form a good data fitting effect, and has the advantages of strong generalization ability, high scalability, and fast operation speed.

  For a dataset with n entries and m dimensions, the XGBoost model can be expressed as follows:

$$\widehat{y} = \sum_{k=1}^{K} f_k(x_i), f_k \in F \tag{16}$$

  Where $F = \left\{f(x) = w_{q(x)}\right\} (q : R^m \to \{1, 2, ..., T\}, w \in R^T)$ Is the set of CART decision tree structures, q is the tree structure from which samples are mapped to leaf nodes, T is the number of leaf nodes, and w is the real number score of leaf nodes.

  When building the XGBoost model, it is necessary to find the optimal parameters according to the principle of minimizing the objective function to establish the optimal model. The objective function of XGBoost model can be divided into error function term L and model complexity function term Ω. The objective function can be written as follows:

$$\begin{cases} Obj = L + \Omega \\ L = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 \\ \Omega = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} {w_j}^2 \end{cases} \tag{17}$$

  When using the training data to optimize the model, it is necessary to keep the original model unchanged and add a new function f to the model to reduce the objective function as much as possible. The specific process is as follows:

$$\widehat{y_i}^{(t)} = \widehat{y_i}^{(t-1)} + f_t(x_i) \tag{18}$$

  In XGBoost algorithm, in order to quickly find the parameters that minimize the objective function, the second-order Taylor expansion of the objective function is performed to obtain the approximate objective function:

To quickly find the parameters that minimize the objective function, the approximation is obtained by performing a second-order Taylor expansion of the objective function, removing the constant term.

If the structural part q of the tree is known, the objective function can be used to find the optimal w_j and obtain the optimal objective function value. Its essence can be attributed to the problem of solving the minimum value of a quadratic function:

$$\begin{cases} {w_j}^* = \frac{-\Sigma_{i \in I_j} gi}{\Sigma_{i \in I_j} h_i + \lambda} \\ Obj = -\frac{1}{2} \sum_{j=1}^{T} \frac{(\Sigma_{i \in I_j} gi)^2}{\Sigma_{i \in I_j} h_i + \lambda} + \gamma T \end{cases} \tag{19}$$

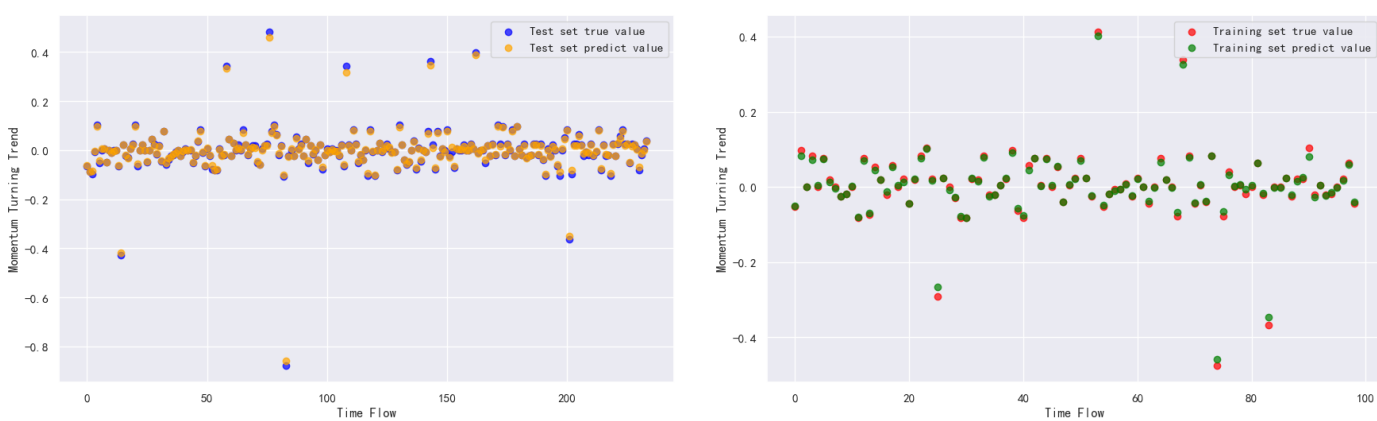
Obj is a scoring function that can be used to evaluate the model, and the smaller Obj is, the better the model is. By recursively building the tree, we set the maximum depth of the tree to 7 and the learning rate to 0.1, expecting an accurate prediction.

- **Test models and Train**

  The match_id "2023-wimbledon-1701" is still utilized for both model training and prediction.

  The first difference of momentum is used as the prediction target, and the time flow is used as the sample size. For the 334 samples in the population, we split it into a test set and a training set in an 7:3 ratio (the first 30% is the train set and the last 70% is the test set).

  Here is a scatter display of the results of model training and prediction:



(a) Testing (b) Training

Figure 6: Train and Test Result Comparison

From above graph we can infer, the model predicts the training and test sets with high accuracy. When the momentum transition trend of players is relatively stable, the prediction can almost accurately judge the momentum transition trend. Even if the prediction of abnormal changes is not particularly perfect, it still has a very high accuracy rate.

- **Horizontal compare results**

  XGBoost is an efficient implementation of GBDT[**5**]. Unlike GBDT, XGBoost adds a regularization term to the loss function. Since some loss functions are difficult to calculate derivatives, XGBoost uses the second-order Taylor expansion of the loss function as the fitting of the loss function.

  In order to test the efficiency and high fit of the XGBoost regression model, we introduce other regression models to make a horizontal comparison. The following table is the evaluation made by the cross-validation set based on the same data set and parameter values:

Table 3: Compare Test Results Table

Model	MSE	RMSE	MAE	MAPE	R^2
XGBoost Regression	0.000	0.006	0.004	0.634	0.996
GBDT Regression	0.002	0.030	0.043	8.373	0.811
CatBoost Regression	0.011	0.107	0.059	7.333	-0.166
LightGBM Regression	0.011	0.108	0.060	14.018	-0.102

  The smaller the MSE (mean square error), RMSE (root mean square error), MAE (Mean Absolute error), and MAPE (mean absolute percentage error) values used in the above table, the higher the accuracy of the model. The r-square compares the predictions to the mean alone, with the closer to 1 the more accurate the model is.

  The R-square of XGBoost test set is 0.996, and the evaluation values of other cross-validation sets are small, and the fitting effect is excellent.

  Compared with GBDT, CatBoost[**6**] and LightGBM[**7**], XGBoost is most suitable for this data validation.

5.3.2 Find Factors most related to Momentum Change

- **Feature Classify**

  When XGBoost regression is used for prediction, it can also be used for feature importance processing. Among them, feature importance is calculated by observing the split number or split gain of the feature in the decision tree.

  Before training the model, we perform feature selection to treat some attributes such as "serve_width" attribute as a class, which has 5 classes: 'W', 'BW', 'BC', 'C', 'B'. We reset these categories to "serve_width_W",..., "serve_width_B" in order to get their corresponding importance scores. After processing the single factors, the corresponding categories were collected to comprehensively analyze the importance arrangement of the influence of the attributes on the momentum transformation trend.

- **Analyze the influence of factors**

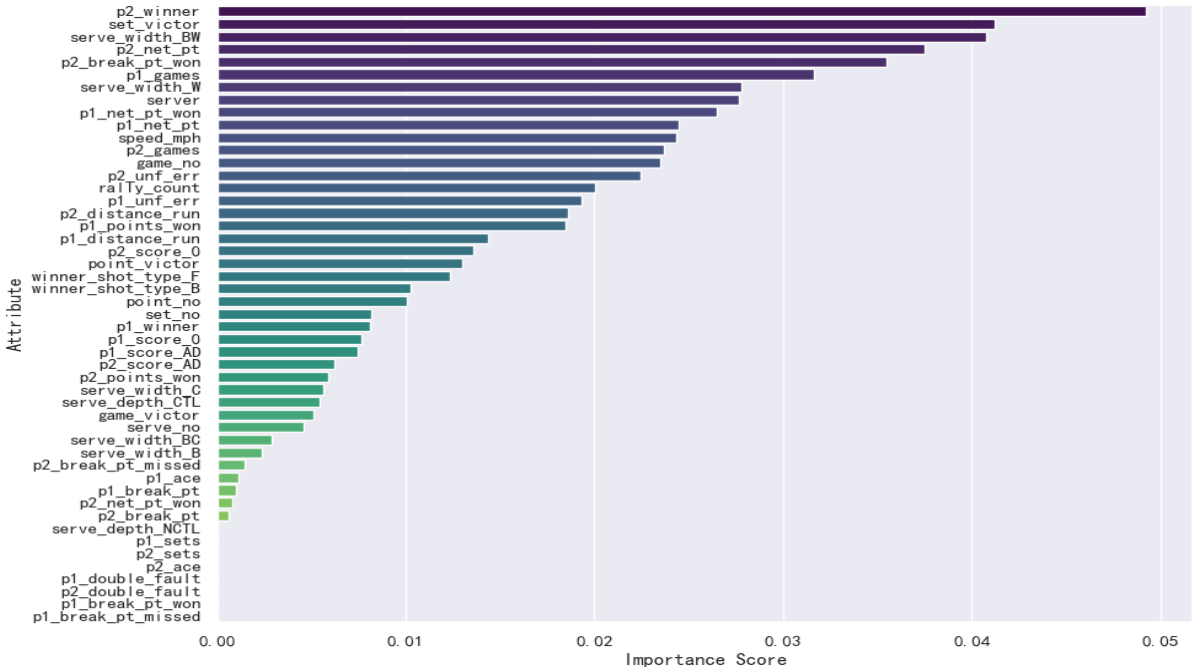


Figure 7: Single Feature Importance

From the provided figures, we can find the factors that affect the players' momentum, which include data such as the scores of the two sides of the match, the serving situation, the type of winner, the width and depth of the court, and the match number.

(1) Maximum Importance: p2_winner (opponent's win) has the highest feature importance, indicating that the outcome of the match is strongly influenced by the opponent's winning score. That is, the opponent's win can easily affect one's momentum. Secondly, set_victor (this point won the set) and serve_width_BW (this point won the set) are also important features, which may increase the probability of the serving side's win and increase the chance of the serving side's win because wider serves are difficult to receive.

(2) Impact of the point score: The score-related features (p1_score_0 and p2_score_AD are just mention here) show varying degrees of influence, reflecting the importance of the different scoring stages in tennis. For example, the scoring status of 0 and AD have different effects on the result of the game, which may be related to factors such as strategy and psychological pressure.

(3) Importance of race progress: The importance of features like set_no, game_no, and point_no indicates that the stage of the race has an impact on the outcome. This may be because the physical and mental abilities of the players change as the game progresses, affecting the outcome.

(4) Player state and ability: The feature importance of p1_distance_run and p2_distance_run indicates that the distance an athlete moves on the field is closely related to the outcome of the game, which may reflect the athlete's physical state and field activity. In addition, unforced errors (p1_unf_err, p2_unf_err) are also important factors affecting the outcome of the game, and reducing errors can significantly improve the probability of winning.

(5) Influence of serve: The feature importance of serve depth (serve_depth) indicates that serve strategy has a significant impact on the outcome of the game. A good serve can create an advantage for the server, and a different service depth may easily result in a winner.

- **Findings in categories**

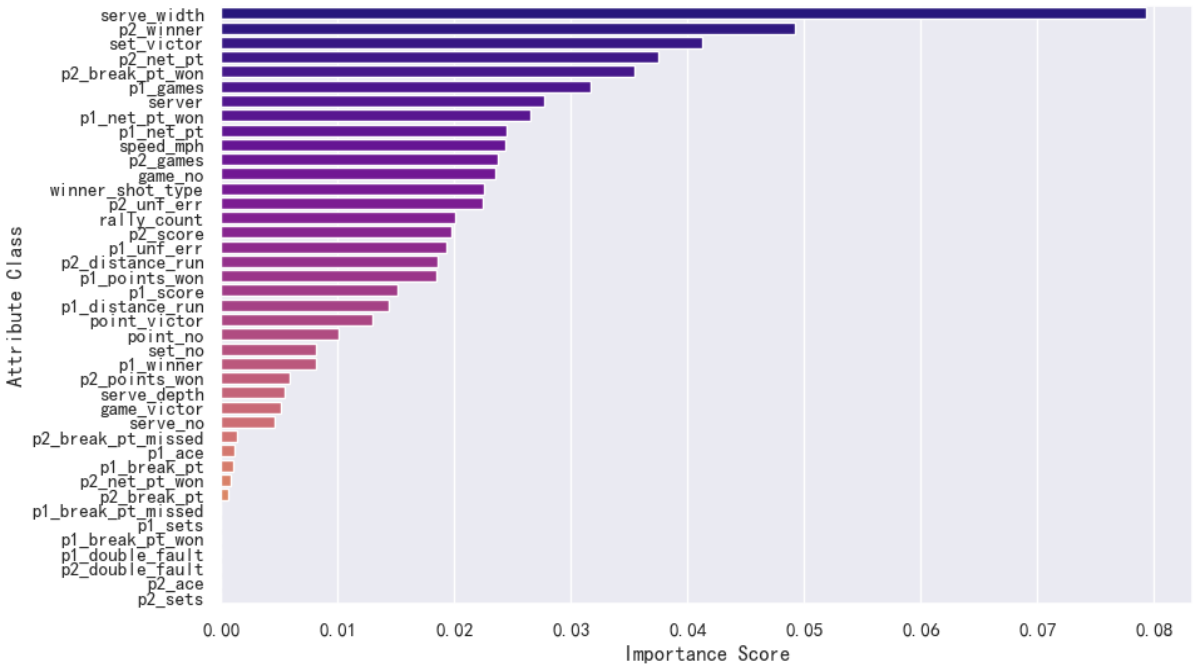


Figure 8: Feature Class Importance

(1) serve_width is one of the most important features with an importance score of 0.079. This shows that precise control of the serve is the key to winning.

(2) The importance score for the opponent's victory (p2_winner) is 0.049. This reflects the tendency of the opposing team to lose momentum when they score.

(3) The feature importance score for set_victor is 0.041, indicating that winning a single set has a significant impact on momentum change or contain.

(4) The feature importance scores of opponent's score in front of the net (p2_net_pt) and opponent's break score (p2_break_pt_won) are 0.038 and 0.035, indicating that performance in front of the net and break success are also very important to the change of the match.

(5) The feature importance scores of player 1's games (p1_games) and server are 0.032 and 0.028, respectively, showing the influence of the accumulation of games and the change of service power on the match.

Overall, these findings highlight the impact of a variety of factors on tennis outcomes, including technique, strategy, and game progression, among others. These characteristics have a great impact on the momentum transfer of the game.

5.4 Question 3 —— Advice to the players

- **Serve:** Serve is one of the most important factors affecting momentum, which includes specific data such as serve depth and serve width, but also important reception such as double faults and breaks of serve. ace is the most desirable form of a player's serve. We recommend that the player emphasize serve speed and accuracy, focusing on improving serve consistency, especially on the second serve. This can be achieved by emphasizing reliability and risk control on the second serve in training. Increase break chances by improving the quality of serve reception and strategy flexibility.

- **Match Points:** One of the nodes where a player's momentum changes the most is around the match point. The pressure on the player rises, both on the opponent's and the player's own match points. When a player wins his match point, his momentum may drop due to slackness; when the opponent wins his match point or the player loses the winning shot, pressure or frustration may also lower his momentum. We recommend that player stay calm on key points, and strengthen their mental training by simulating stressful situations in matches to improve their mental toughness during matches. Staying focused during the game, effectively managing stress and emotions, maintaining the habit of observing and analyzing the opponent's performance, and timely adjusting the game plan to cope with different game situations.

- **Initiative:** The player's direct score, the player's running distance and fewer unforced errors can make the player's momentum rise quickly, on the contrary, too many unforced errors will cause the player's momentum to fall very quickly. We recommend that the player improves baseline aggressiveness, utilizes every opportunity to create direct scoring, improves control of power, angle and depth, improves control of the ball and avoids too many errors. Focus on improving your ability to judge the ball and select more sensible strokes.

5.5 Question 4

5.5.1 Sensitivity analysis and test other match

In order to test the stability and accuracy of the above model, we choose another match "2023-wimbledon-1301". Then we test this data and get the result below:

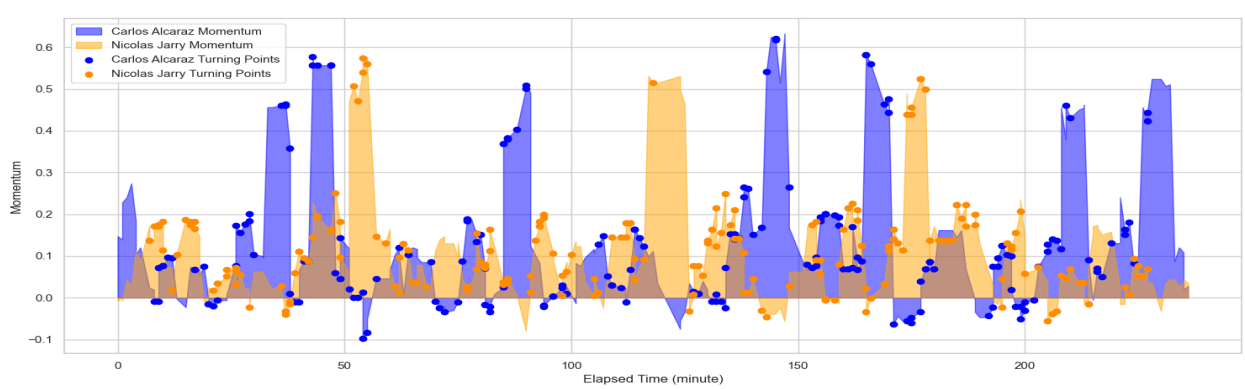
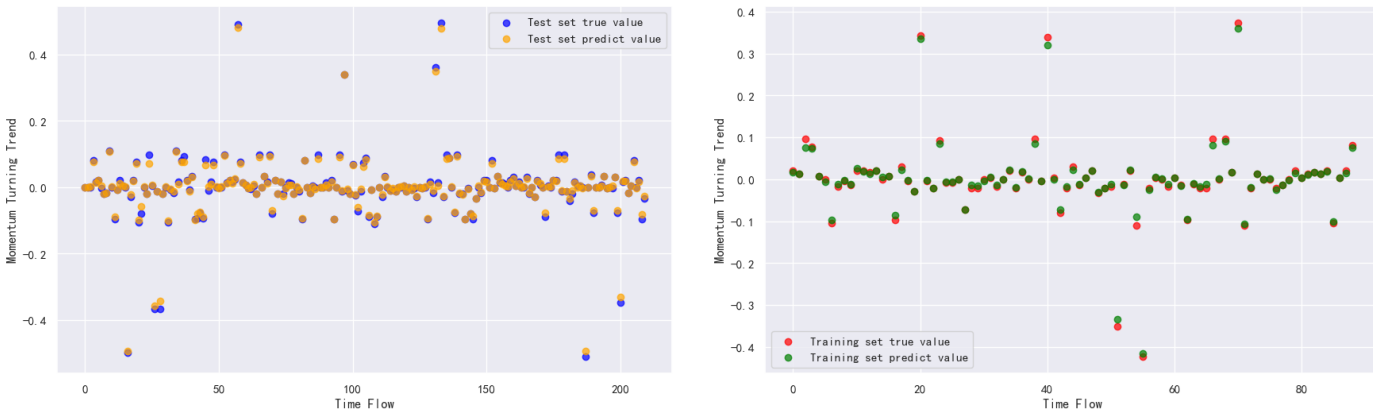


Figure 9: Momentum Change with Turning point in "2023-wimbledon-1301" match

It can be seen that in this test at the trend of the momentum value changes, the model can be correctly displayed at the change point, indicating that the model has a high accuracy.

- **XGBoost Regression Model**



(a) Test (b) Train

Figure 10: Train and Test Result Comparison

Table 4: XGBoost Regression Test Results Table

MSE	RMSE	MAE	MAPE	R^2
0.001	0.005	0.004	0.878	0.995

Using player 1 as the baseline, we get the above data. The prediction can almost accurately judge the momentum transition trend. It has a very high accuracy. The R-square of XGBoost test set is 0.995, and the evaluation values of other cross-validation sets are small, and the fitting effect is excellent.

- **Model test results:** The prediction of swings in matches is very accurate, with an accuracy of more than 98%. It indicates that the model is stable and highly correct.

5.5.2 Factors that may need to be included in future improvements to the model

If the model performs poorly in some cases, it may mean that there are some factors that are not fully considered by the current model. In order to improve the accuracy and generalization ability of the model, consider including the following factors in future models:

1. Players' own factors

Physical condition: A player's decreased physical condition may result in imprecise technical movements and decreased tactical execution.

Psychological state: The player's stress management, psychological adjustment ability has an important impact on the result of the game.

Injuries: The fitness of the players and whether injuries may affect their performance.

2. Environmental and external factors

Court surface: Different court surfaces can significantly affect the bouncing characteristics of the ball and the rhythm of the game.

Climatic conditions: Climatic conditions such as temperature, humidity, wind speed can also have an impact on player performance and the flight path of the ball.

Crowd support: Home advantage and crowd support may have an impact on a player's state of mind and performance.

3. Game-related factors

Tournament stage and importance: such as Grand Slam, regular tour, knockout tournament may affect the strategy and mental state of the players.

5.5.3 Generality of the model

For other tournaments, the model may need to be adjusted for different types of tournament characteristics (e.g. women's tournaments, different levels of tournaments), as these tournaments may have different swings and influencing factors.

In big tournaments such as Grand Slams, men's matches are usually best-of-five sets, while women's matches are best-of-three[**9**]. This means that women's matches are usually shorter and end more quickly.

If the surface area of the field is large, the distance and difficulty of the player's movement will be increased, and the smoothness of the ground will also have a corresponding impact. This problem is not yet matched by our model. However, we can add venue attributes to this data and use them in the model for solving.

For other ball games, such as table tennis, the 11-point scoring system is used[**10**], that is, a player needs to reach 11 points and lead his opponent by two points to win a game. And table tennis matches are usually short, each game may only take a few minutes. The length of tennis matches, especially five-set men's matches in big tournaments, can take hours or more. So the momentum shift in table tennis may be more densely distributed.

Therefore, we need to change the values of the attributes corresponding to the input in the model, but the basic parameters of the model do not change, because the above model processing of the attributes is based on the data itself, so the model is still very adaptable to different competition situations. The generalization of our model can deal with the above problems.

6 Model Evaluation

6.1 Strength

- **Simple to understand:** The principle is relatively simple, easy to understand, and implement. It does not require complex calculations or algorithms and is suitable for all levels of users.
- **The universality of application:** It can be applied to various decision-making problems

and data-processing problems, including evaluation, ranking, selection, etc. It applies to a variety of fields and industries, such as environmental assessment, investment decision-making, talent selection, and so on.

- **Objectivity:** The influence of subjective factors on the results is avoided. It can analyze and model according to the index objectively.
- **High efficiency:** The data can be processed and analyzed clearly, and clear conclusions can be obtained.

6.2 Weakness

- **Ignore empirical judgment:** It mainly relies on data analysis and ignores empirical judgment. In some cases, experience can play an important role.
- **Computational complexity:** In the face of some complex and cumbersome data, the model needs to be optimized.
- **High data requirements:** During the Sliding window model, some data will be discarded. This can result in some critical data being ignored or lost, and If there is a large error or missing of the indicator data, the calculation result may be inaccurate.

7 Memo: Conclusion & Advice for Coaches

7.1 Conclusion

- **Momentum detection**

  When studying momentum and performance over a provided period of time, the factors that influence momentum can include the following:

  Scoring Advantage: This compares the difference in scores between two players over a specific period of time. The higher the difference, the higher the score, indicating that the player has an advantage.

  Games Won: This compares the number of games won by each player over a period of time. A higher number of times indicates that the player has a greater momentum.

  Server: In tennis, the server has a great advantage over the player. We consider the fraction of the points won by the serving player and whether or not the serving player is able to hold his or her service game (i.e. win his or her service game).

  BreakPoint: A breakpoint indicates whether a player can win the match while his opponent is serving.

  Unforced Errors: Fewer unforced errors can be seen as better performance, and conversely, more unforced errors can be seen as worse performance.

- **Randomness for swings in play and runs of success**

  Swing and success of a player in a match are not random, but have a certain positive correlation with Momentum.

- **Factors most related flow of play changing**

  The influencing factors of game swings can be briefly summarized as follows:

  The score affected the results to varying degrees, with different effects in different scoring stages, possibly related to strategy and pressure.

  An opponent's victory may affect one's own motivation. At the same time, winning a set and serving width are also important.

  The service strategy has an important impact on the outcome of the match, and a good serve can create an advantage for the server.

  The stage of the progression affects the final outcome, as the level of physical exhaustion and mental state of the players change as the game progresses, thus affecting the outcome.

  The ability of the player has an impact on the outcome of the match, such as the distance the player moves on the court and unforced errors, and reducing errors can significantly improve the probability of winning.

7.2 Suggestion for Coaches

From: Team #2403861

To: Tennis Coaches

These are some of our conclusions and recommendations:

- Training players not only to train their service skills but also should train receiving skills, and multi-faceted development, for the key role of service in determining the outcome of the game and scoring performance.
- In the interval of the match, fully understand the personal characteristics and playing style of the player, and help the player seize the breakpoint opportunity because the player who successfully grabs the breakpoint opportunity at the key moment can usually gain the advantage of the match and finally win. This shows that breakpoint chances play an important role in determining the outcome of the match.
- Facing the impact of unforced errors on players, studies have shown that there is a significant correlation between the number and frequency of unforced errors and match results. Specifically, an increase in unforced errors tends to be accompanied by a decrease in match-winning percentage, while a decrease in unforced errors contributes to an increase in match-winning percentage. Therefore, for players who are often affected by unforced errors, daily training should be more solid and stable, and at the same time, more attention should be paid to the mental health of players, and the mentality of players should be cultivated.

Yours,
Sincerely,
Team #2403861 of MCM 2024

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Report on Use of AI

1 OpenAI ChatGPT (2024 version, ChatGPT-3.5)

Query1: match_id p1_momentumchange p2_momentumchange p1_turning_pointschange p2_turning_pointchange 1 2023-wimbledon-1701 1 1 0 0

Output: These are four columns of data in a data frame, with each row representing a tennis match. match_id: A unique identifier for the tournament. p1_momentumchange: Indicates whether the momentum of athlete 1 (p1) is random, 1 means it is random, 0 means it is not. p2_momentumchange: Indicates whether the momentum of athlete 2 (p2) is random, 1 means it is random, 0 means it is not. p1_turning_points:change Indicates whether the turning point of player 1 (p1) is random or not, where 1 is random or 0 is not. p2_turning_pointscahnge: Indicates whether the turning point of player 2 (p2) is random or not; 1 means it is random or 0 means it is not. These data may be the results obtained by calculating and analyzing the score data of each game and are used to study the performance and dynamics of the players during the game.

2 OpenAI ChatGPT (2024 version, ChatGPT-3.5)

Query1: XGBoost does feature importance processing

Output: Here are the general steps for feature importance processing with XGBoost: Prepare the data: Divide the dataset into training and test sets and do the necessary preprocessing (like missing values handling, feature encoding, etc.). Define Model: Import the XGBoost library and define an XGBoost model. Depending on the problem type, you can choose a regression model (xgboost.XGBRegressor) or a classification model (xgboost.XGBClassifier). Train model: Train the XGBoost model using the training set. Various hyperparameters can be set such as learning rate, number of trees, maximum depth, etc. Feature Importance calculation: With the trained model, the feature importance score can be obtained using the feature_importances_ attribute. This score indicates how important each feature is in the model.